

Sonal Gehlot

sonalgehlot10@gmail.com | +1 (206) 746-9460 | [linkedin.com/in/sonalgehlot/](https://www.linkedin.com/in/sonalgehlot/) | sonalgehlot10.github.io

EXPERIENCE

Amazon Web Services

Software Development Engineer Intern

Jun 2026 – Aug 2026

- Developed a self-service, full-stack, multi-region AWS test environment provisioned entirely within each developer's own account (via CDK), eliminating reliance on a shared testing stage for a 20-engineer team and cutting iteration time from 4 hours to 10 minutes (24x faster).
- Updated CDK infrastructure across all services to support multi-region resource deployment within a single AWS account, including cross-region DNS resolution keyed to each developer's alias, enabling realistic multi-region testing without emulation.
- Created a one-click script that fully provisions the multi-region test environment for 8+ onboarded services in a developer's AWS account. Authored an agent skill to onboard new services onto the environment and debug setup failures in existing services.

SIMULABS

Jr Full Stack Developer

Aug 2024 – Dec 2024

- Worked on two full-stack MERN SaaS applications with Firebase Auth and OpenAI API integrations, serving GenAI features to end users.
- Designed REST APIs and automated CI/CD deployment pipelines, cutting release cycle times by 40% and improving overall service uptime.

Sparks Foundation

Web Developer / Designer

Feb 2023 – May 2023

- Rebuilt backend and payment gateway UX with modular APIs and validation logic, reducing load times by 20%.

Yoganta, TakeOff

Video Editor

Aug 2021 – Feb 2023

- Led 30+ video and design projects, improving brand engagement by 25% through creative automation.

Verzeo

Software Engineer Intern

Mar 2021 – Apr 2021

- Built full-stack ML web app (ReactJS, MongoDB) and reduced latency 30% via caching and async processing.

PROJECTS & PUBLICATION

Distributed Key-Value Store using Raft Consensus | C++, gRPC, Protocol Buffers, Distributed Systems

- Built a fault-tolerant C++/gRPC key-value store using Raft consensus, achieving linearizable reads/writes across replicated nodes.
- Implemented leader election, log replication, and heartbeat mechanisms to handle concurrent client requests and network partitions.

RLVR: Reducing Hallucinations in LLMs | Python, PyTorch, HF TRL, RLHF/RLVR, Gemma-3B, Qwen-4B

- Developed and published an RL training pipeline that replaces human feedback with verifiable reward signals to reduce LLM hallucinations.
- Fine-tuned 3B–4B models (Gemma, Granite, Qwen), achieving 21% controlled abstention on large-scale QA benchmarks without degrading accuracy. **arXiv:** arxiv.org/abs/2601.20126 (Jan 2026)

Breast Cancer Detection Using ML | Naïve Bayes, KNN, scikit-learn, NumPy, Pandas, PHP, MySQL, XAMPP

- Designed a full-stack predictive healthcare web app (PHP, MySQL, XAMPP) that classifies breast cancer as benign or malignant using Naïve Bayes and KNN (scikit-learn) on the WBCD dataset, with custom backend integration for real-time ML inference. **Published in IRJET:** [irjet.net/archives/V9/i3/IRJET-V9I3296.pdf](https://www.irjet.net/archives/V9/i3/IRJET-V9I3296.pdf) (Mar 2022)

EDUCATION

University of Southern California

Jan 2025 – Dec 2026

M.S. in Computer Science (GPA: 3.77/4.0)

Los Angeles, CA

- Coursework:** Distributed Systems, Analysis of Algorithms, Database Systems, Machine Learning for Data Science, Applied NLP

University of Mumbai

Jun 2018 – May 2022

B.E. in Information Technology (CGPA: 8.85/10)

Mumbai, India

TECHNICAL SKILLS

Proficient: AWS Services, CDK (Infra-as-Code), Python, Java, REST APIs, Git

Familiar: C++, TypeScript, JavaScript, PyTorch, Firebase, CI/CD, MongoDB, MySQL